

7.1 The Pythagorean Theorem

Lesson Introduction

 Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles. MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.		 Learning Target	I can find the missing side of a right triangle using the Pythagorean Theorem.
 Benchmark Coherence	Building On MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.		Working Toward N/A	
 Essential Questions	- How do you find the missing side of a right triangle? - What is the relationship between the sides of a right triangle?			
 Lesson Narrative	This lesson allows students to work together to explore the relationship between the sides of a right triangle. Using the information found in the Collaboration, students will work through practice problems over the Pythagorean Theorem. Students will then complete Guided Practice problems and individual practice over Pythagorean Theorem.			
 Component Summary	7.1.1 Warm-Up (~5 min.)	Discussing success from failure as told by a Sales Strategist (<i>SE p. 369</i>)	7.1.3 Guided Practice (15-20 min.)	Solving for missing sides of a right triangle (<i>SE p. 372</i>)
	7.1.2 Collaboration (10-15 min.)	Exploring the Pythagorean Theorem (<i>SE p. 370</i>)	7.1.4 Your Turn! (~10-15 min.)	Applying Pythagorean Theorem to find the missing side length of a right triangle (<i>SE p. 373</i>)
	7.1.5 Wrap-up (~5 min.)	Students summarize their learning (<i>SE p. 374</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Hypotenuse - Pythagorean Theorem		- Failure to Find Success Video - Graphing utility (Optional): Index cards or copy paper, colored pencils, scissors	
		Additional Preparation 7.1.1 Warm-Up contains a video to accompany the question prompts		

 Teacher Tips	Common Misconceptions - Students may struggle with differentiating between the legs and hypotenuse of right triangles. - Students may believe the hypotenuse, or c, will always be the missing side length, still finding the sum of a and c rather than the difference in those cases.	Things to Consider Consider allowing students to cut out copies of the squares in 7.1.2 Collaboration to find the connection between the area of the squares forming the legs, and the area of the square forming the hypotenuse.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder At the start of the Collaboration during 7.1.2 Collaboration, have students discuss what they notice and wonder. This allows students to begin asking questions and exploring the Pythagorean Theorem before working with the Pythagorean Theorem, making informal observations that ultimately connect to formalized concepts.	MA.K12.MTR.3.1: Mathematical Fluency Allow students multiple opportunities to solve Pythagorean Theorem problems that are presented in different ways. Allow students time to discuss their way of solving with a partner and provide feedback for students.
 Differentiate Instruction	In 7.1.2 Collaboration, students have the opportunity to explore one real-world application of the Pythagorean Theorem by investigating the shorter distance between two different walking paths from the beach. Consider this scenario as an opportunity to differentiate by content to engage your students in ways unique to your contexts. - Provide an example from your school campus or local community that students would be familiar with and may agree or disagree based on their personal experience. - Prompt students to create their own example based on a location they are familiar with or walk often. Challenge them to find actual distances before or after class to investigate or verify their findings mathematically.	
Lesson Notes:		

7.1.1 Warm-Up: Sales Strategist

Component Overview: Students will look at how to respond to failure after watching a short video.



7.1.1 Warm-Up: Sales Strategist

Kishan Patel says that he was never really a good student in high school. He graduated with a 2.9 GPA. All of those challenges prepared him for business. Business is about failing and doing better. Every time you fail, you have to improve in some way to get to the next level.

1. Based on the statement above, how do you think Kishan views failure?

Answers will vary.

2. When you fail at something, describe how you typically react.

Answers will vary.



Consider allowing a few students to share their responses. Do not spend too much time on this topic, but instead allow it to be used to explore Growth Mindset.

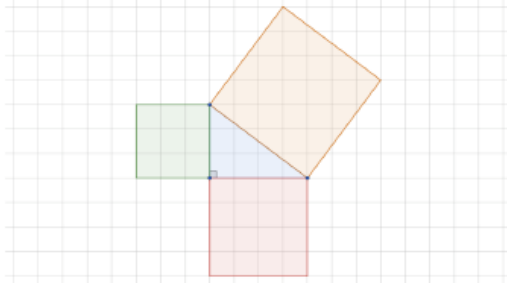
7.1.2 Collaboration: Pythagorean Theorem

Component Overview: Students will work in pairs or groups to explore the relationship between the legs and hypotenuse of a right triangle.



7.1.2 Collaboration: Pythagorean Theorem

A right triangle is shown with three squares.



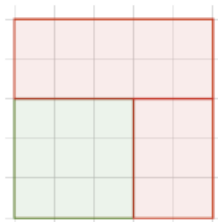
- The three squares from the diagram are shown. Complete the table with the area of each square.



Side Length (units)	Area of Square (square units)
3	9
4	16
5	25

- Explain how the diagram shows that the area of the green square plus the area of the red square is equal to the area of the orange square.

The number of unit squares in both the green and red squares make up exactly the same number of unit squares in the orange square.



For a kinesthetic activity, have students use paper or index cards to cut out each square for this activity. Students can use the cut outs to explore the relationship between the orange square and the red and green squares.



Notice and Wonder

Before students answer questions, consider using this instructional routine to allow students to write down things they notice and wonder about the image.

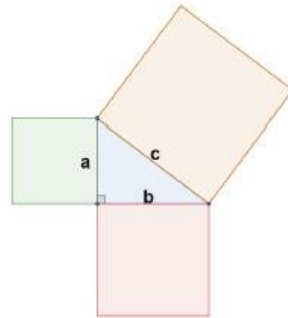
- What do you notice about the image?
- What do you wonder about the image?

7.1.2 Collaboration: Pythagorean Theorem (continued)

3. Complete the relationship of the areas by filling in the box.

$$3^2 + 4^2 \boxed{=} 5^2$$

4. Use the diagram to write an explanation of the Pythagorean Theorem, $a^2 + b^2 = c^2$.
Sides a and b are legs of the right triangle. Side c is the hypotenuse. If the lengths of these two sides, a and b , are squared and then added together, their sum will be equal to the length of the hypotenuse, c , squared.



5. Iverson and Jake are at the beach and plan to meet a friend for lunch.



Iverson decided to walk directly to the restaurant, while Jake decides to take a route that will allow him to cross the street using the crosswalk. The diagram shows the two routes. Jake walks horizontally 252 meters (m) and then vertically 115 m.



Think-Pair-Share

Provide students an opportunity to share their group discussions with other groups in order to discuss different opinions.



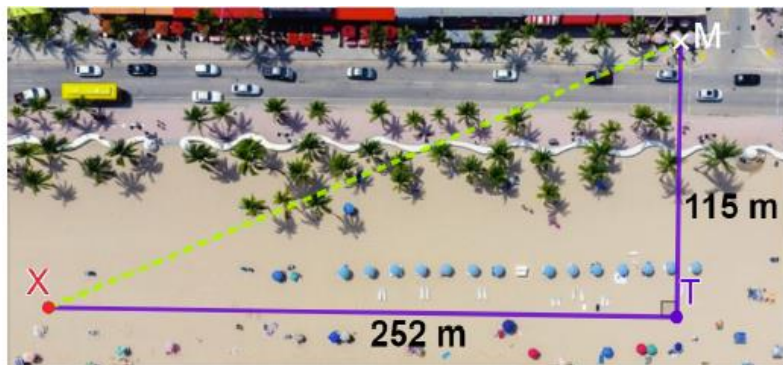
Utilize inquiry for students to make connections between the visual squares and the Pythagorean Theorem:

- What is the relationship between the green/red squares and the orange square?
- Would that always be true? Verify.
- What does that make you think about the relationship between the two legs of a right triangle and their hypotenuse?



Students who missed the 8th grade benchmarks may not know the definition of hypotenuse.

7.1.2 Collaboration: Pythagorean Theorem (continued)



a. Complete the table.

	Iverson	Jake
Distance Walked	277 m	367 m

b. Discuss with your partner the advantages and disadvantages of each person's walk to the restaurant. Summarize your discussion in the table.

Answers may vary.

Person	Advantages	Disadvantages
Iverson	<i>Shorter distance.</i>	<i>Not using the crosswalk which is illegal to cross a road.</i>
Jake	<i>Using the crosswalk which is the legal way to go across a street.</i>	<i>Longer distance.</i>



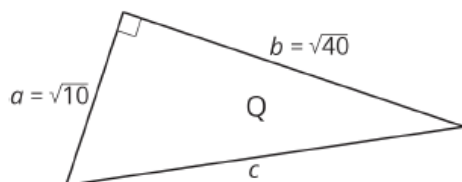
Which way do you think is the quickest way to get to the lunch spot?



When would it be better to take the longer path versus the direct path?

7.1.2 Collaboration: Pythagorean Theorem (continued)

6. A right triangle is shown with side lengths $\sqrt{10}$ and $\sqrt{40}$ units.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

- a. State Pythagorean Theorem

$$a^2 + b^2 = c^2$$

- b. Substitute

$$\sqrt{10}^2 + \sqrt{40}^2 = c^2$$

- c. Solve for the missing variable.

$$10 + 40 = c^2$$

- d. Solutions

$$50 = c^2$$

Exact solution:

$$c = \sqrt{50}$$

Approximate answer:

$$c \approx 7.071$$



- Why does $\sqrt{277^2} = 277$, and $\sqrt{367^2} = 367$?
- What does that make you think about what your final answer for the hypotenuse might look like?

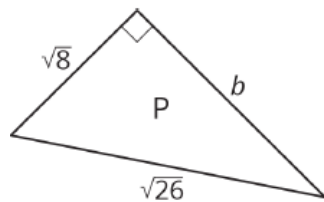
7.1.3 Guided Practice: Pythagorean Theorem

Component Overview: Students use the Pythagorean Theorem to find the missing side lengths of right triangles.



7.1.3 Guided Practice: Pythagorean Theorem

- Find the exact measure of b in the diagram.



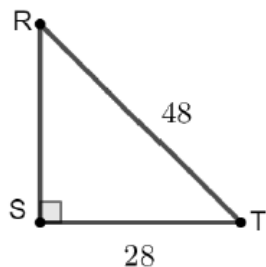
$$\sqrt{8}^2 + b^2 = \sqrt{26}^2$$

$$8 + b^2 = 26$$

$$b^2 = 18$$

$$b = \sqrt{18}$$

- $\triangle RST$ is shown, where $ST = 28$ and $RT = 48$.



Find the length of $\overline{SR}$ to the nearest thousandth.

$$(SR)^2 + 28^2 = 48^2$$

$$(SR)^2 + 784 = 2304$$

$$(SR)^2 = 1520$$

$$SR = 38.987$$



Use observation data from the 7.1.2 Collaboration to determine how much guidance is needed to bring together key ideas in this Guided Practice. If necessary, consider modeling question 1 to address common misconceptions shown in the Collaboration.



Students may think the missing side is always the c -value in the equation. Consider having students draw a line or arrow from the right angle pointing towards the hypotenuse, in order to find c .



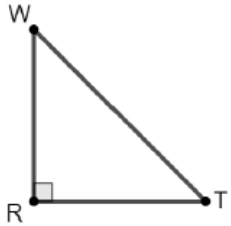
In question 2, students may find that $SR = 20$ if they subtract 48 and 28 before squaring. Facilitate a quick discussion about how to use order of operations when solving an equation.



Consider allowing students to use a calculator for question 2 and question 3. Determine what place value you would like your students to round to for their final answer.

7.1.3 Guided Practice: Pythagorean Theorem (continued)

3. $\triangle WRT$ is shown, where $WT = 18.4$ and $RT = 12.1$.



Find the length, to the nearest thousandth, of $\overline{WR}$.

$$(WR)^2 + 12.1^2 = 18.4^2$$

$$WR = 13.862$$



This problem provides an opportunity to determine a student's understanding of solving algebraic equations involving squares and square roots.

Students who struggle with radicals would benefit from a review of solving equations using Algebra 1 On-Ramp "Simplifying Square Roots," and/or "Approximating Irrational Numbers."

7.1.4 Your Turn!

Component Overview: Students practice using the Pythagorean Theorem to find the missing side lengths of right triangles.



7.1.4 Your Turn!

1. In $\triangle HMX$, $m\angle MXH = 90^\circ$, $MH = 12$, and $MX = 7$. Determine the length of $\overline{XH}$. Round to the nearest thousandth.

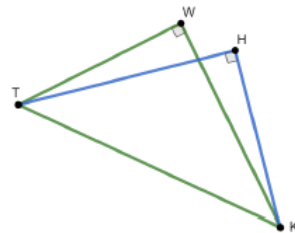
$$7^2 + (XH)^2 = 12^2$$

$$49 + (XH)^2 = 144$$

$$(XH)^2 = 95$$

$$XH = 9.747$$

2. $\triangle TWK$ and $\triangle THK$ are shown, where $TW = \sqrt{45}$, $TH = \sqrt{68}$, and $TK = \sqrt{113}$. Find the length of $\overline{WK}$ and $\overline{HK}$. Round to the nearest thousandth.



$$(WK)^2 + \sqrt{45}^2 = \sqrt{113}^2$$

$$WK = 8.246$$

$$(HK)^2 + \sqrt{68}^2 = \sqrt{113}^2$$

$$HK = 6.708$$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

7.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about the Pythagorean Theorem.

Answers will vary.